

WEEKLY PROGRESS REPORT – WEEK 5

Project Title

Healthcare Analytics Pipeline for MediCare Hospital

Group Number: 3

Team Members

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Week 5 Objectives

The objective of Week 5 was to perform predictive analytics, develop visualization dashboards, complete project integration, and prepare final project deliverables.

Activities Completed

1. Machine Learning Implementation

A Random Forest model was developed to predict patient Length of Stay using healthcare attributes and historical patient records.

2. Model Evaluation

The machine learning model was evaluated using Mean Absolute Error (MAE) and performance metrics were documented.

3. Streamlit Dashboard Development

An interactive Streamlit dashboard was developed to visualize healthcare KPIs, disease trends, treatment costs, and patient analytics.

4. Apache Kafka Demonstration

Apache Kafka Producer and Consumer workflows were implemented to demonstrate real-time healthcare message streaming capabilities.

5. GitHub Actions CI/CD Automation

A GitHub Actions workflow was configured to demonstrate Continuous Integration and Continuous Deployment practices.

6. Final Testing and Validation

All project components were tested and validated, including ETL workflows, analytics, machine learning outputs, dashboard functionality, and cloud integrations.

7. Documentation and Presentation Preparation

The final report, architecture diagram, GitHub repository documentation, presentation slides, and project demonstration materials were completed.

8. Team Coordination and Final Review

The team reviewed all project deliverables, validated outputs, and finalized the project for submission.

Deliverables Completed

- Machine learning model
- Evaluation report
- Streamlit dashboard
- Apache Kafka demonstration
- GitHub Actions CI/CD workflow
- Final report
- Presentation slides
- Project documentation
- Demo preparation

Challenges Faced

- Integrating multiple technologies into a single workflow
- Validating end-to-end pipeline execution
- Testing dashboard functionality
- Preparing project documentation and presentation materials

Plan for Final Submission

- Submit final report
- Submit presentation

- Submit GitHub repository
- Submit dashboard link
- Submit project demonstration video

Conclusion

Week 5 focused on machine learning, dashboard development, project integration, CI/CD automation, Kafka streaming demonstration, and final documentation. The Healthcare Analytics Pipeline was successfully completed as an end-to-end data engineering solution for healthcare analytics.